

Lean 4 formalization of the Kontsevich-Segal axioms

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Chapter 1

The domain of complex metrics (KS Section 2)

This chapter is the blueprint for Section 2 of Kontsevich–Segal, *Wick rotation and the positivity of energy in quantum field theory* (arXiv:2105.10161), formalized in `KontsevichSegal/ComplexMetrics/`. The paper introduces, for a d -dimensional real vector space V , an open domain $Q_{\mathbb{C}}(V)$ of *allowable* complex-valued quadratic forms (“complex metrics”), generalizing positive-definite inner products, whose Shilov boundary contains the real Lorentzian metrics — and no other nondegenerate real metrics.

Two project conventions (see `CLAUDE.md`) govern this blueprint. First, the *working definition* of allowability in the Lean code is the diagonal characterization of Theorem 3 (the angle condition), because Definition 1 requires the Hodge star operator, which Mathlib lacks. Second, the *no-approximations policy*: each blueprint node below states the paper’s actual claim, never a weakened version; nodes whose faithful statement is not yet expressible in Lean carry a `True` placeholder on the Lean side and are marked not ready here, with the blocking infrastructure named.

1.1 Allowable complex metrics

Definition 1 (Allowable complex metric, Hodge-star form; KS Definition 2.1). On a d -dimensional real vector space V , a quadratic form $g: V \rightarrow \mathbb{C}$ is an *allowable complex metric* if, for all degrees $p \geq 0$, the real part of the quadratic form

$$\Lambda^p(V^*) \longrightarrow |\Lambda^d(V^*)|_{\mathbb{C}}, \quad \alpha \mapsto \alpha \wedge *_g \alpha$$

is positive-definite. Here $*_g$ is the Hodge star operator of the complex metric g , and $|\Lambda^d(V^*)|_{\mathbb{C}}$ is the complexified twisted determinant line, with orientation-reversing automorphisms of V acting antilinearly.

Not formalized. There is no Lean counterpart of this definition: the Hodge star operator for (complex) metrics, and the induced quadratic forms on the exterior powers $\Lambda^p(V^*)$, are not available in Mathlib. The Lean development instead takes Definition 2 below as the working definition.

Definition 2 (Allowable complex metric, working definition: the angle condition). A tuple of complex numbers $(\lambda_1, \dots, \lambda_d)$ satisfies the *angle condition* if every λ_i is nonzero, no λ_i lies on the

non-positive real axis, and

$$|\arg(\lambda_1)| + |\arg(\lambda_2)| + \cdots + |\arg(\lambda_d)| < \pi. \quad (4)$$

An *allowable complex metric* on a finite-dimensional real vector space V is a symmetric nondegenerate $\mathbb{R}$ -bilinear form $g: V \times V \rightarrow \mathbb{C}$ for which there exist a basis (y_i) of coordinates on V and coefficients (λ_i) satisfying the angle condition with

$$g = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_d y_d^2.$$

The collection of allowable complex metrics on V is denoted $Q_{\mathbb{C}}(V)$; in the paper it is an open subset of the complex vector space $S^2(V_{\mathbb{C}}^*)$.

In Lean: **AngleCondition** (the three conditions on the coefficients), **AllowableComplexMetric** (the bilinear form with symmetry, nondegeneracy and the diagonal characterization), and **QC** (the domain, as a type). All three are fully defined.

Theorem 3 (Equivalence of the two definitions; KS Theorem 2.2). *A quadratic form $g: V \rightarrow \mathbb{C}$ satisfies Definition 1 if and only if it can be written, in some basis of V , as $\sum_i \lambda_i y_i^2$ with nonzero coefficients λ_i , not on the negative real axis, satisfying the angle condition (4) of Definition 2.*

Lean status: **True** placeholder. *The biconditional cannot be stated because the Hodge-star side (Definition 1) is not expressible; blocked on Hodge star / induced forms on exterior powers in Mathlib.*

Proof. (Paper, p. 10.) A form satisfying Definition 1 with $p = 1$ has $\operatorname{Re}((\det g)^{-1/2} g)$ positive-definite, which allows simultaneous diagonalization of the real and imaginary parts of g over $\mathbb{R}$. In a diagonalizing basis the form $\alpha \mapsto \alpha \wedge *_g \alpha$ on $\Lambda^p(V^*)$ is diagonal with values $(\lambda_1 \cdots \lambda_d)^{1/2} \prod_{i \in S} \lambda_i^{-1}$ on the basis p -forms e_S^* ; positivity of all their real parts for every subset S is precisely condition (4). $\square$

Lemma 4 (Values form a convex cone; KS p. 9). *For $g \in Q_{\mathbb{C}}(V)$ one has $\max_i \arg \lambda_i - \min_i \arg \lambda_i < \pi$, so as v runs through V the values $g(v)$ form a closed convex cone in $\mathbb{C}$ disjoint from the open negative real axis. In particular $g(v)$ is never real and negative.*

Lean status: sorry. *The declaration `not_neg_real_axis` states the “in particular” clause (for $v \neq 0$, $g(v, v)$ is not real and negative); the convex-cone clause itself is not yet stated in Lean.*

Proof. From (4), if $\arg \lambda_j > 0 > \arg \lambda_k$ then $\arg \lambda_j - \arg \lambda_k \leq \sum_i |\arg \lambda_i| < \pi$; hence all values $\lambda_i y_i^2$ lie in a closed convex cone of opening angle less than π , which contains no two opposite rays and avoids the open negative real axis. Nonnegative combinations stay in the cone. $\square$

Lemma 5 (The volume element; KS p. 7). *For $g \in Q_{\mathbb{C}}(V)$ the determinant $\det g$ (of the Gram matrix in any basis; it is invariantly defined up to multiplication by a positive real number) is not real and negative, and the square root $(\det g)^{1/2}$ can be chosen with strictly positive real part. This fixes the holomorphic volume element $\operatorname{vol}_g = (\det g)^{1/2} |dy^1 \cdots dy^d|$.*

Lean status: sorry (`volume_element_positive`).

Proof. In a diagonalizing basis $\det g = \lambda_1 \cdots \lambda_d$, whose argument is $\sum_i \arg \lambda_i \in (-\pi, \pi)$ by (4); a base change multiplies the Gram determinant by $(\det P)^2 > 0$. Since $\arg \det g \in (-\pi, \pi)$, the principal square root has $|\arg| < \pi/2$, i.e. positive real part. $\square$

Proposition 6 (Configurations in the holomorphic envelope; KS Proposition 2.3). *If a complex metric on a d -dimensional real vector space V satisfies condition (4), then the configuration space $\operatorname{Conf}_k(V)$ of k -tuples of distinct points of V is contained in the holomorphic envelope of the Wightman permuted extended tube $\mathcal{U}_k(V) \subset (V_{\mathbb{C}})^k$.*

Not formalized (deferred). *Requires the Wightman permuted extended tube, holomorphic envelopes of domains in $(\mathbb{M}_{\mathbb{C}})^k$, and Bochner's tube theorem — none of which are formalized. Planned for the WickRotation/ phase.*

Proof. (Paper, appendix to Section 2.) Identify V with $\exp(i\Theta/2)(\mathbb{E}) \subset \mathbb{M}_{\mathbb{C}}$ where Θ lies in the generalized octahedron Π_0 cut out by (4). One shows $\exp(i\Theta/2)(\mathbf{x}) \in \mathcal{U}_k$ when $\text{Re}(\Theta)$ lies on a coordinate axis (Π_{00}), then extends to the convex hull Π_0 of Π_{00} by Bochner's tube theorem; the reduction of the $d = 2$ case to the standard tube theorem is Lemma 13. $\square$

1.2 Structure of the domain $Q_{\mathbb{C}}(V)$

Definition 7 (The fiber $\Pi(V)$: trace-norm ball; KS p. 11). For a positive-definite inner product g_0 on V , let $\Pi(V, g_0)$ denote the convex open set of g_0 -self-adjoint operators $\Theta: V \rightarrow V$ with trace-norm $\|\Theta\|_1 = \sum_i |\theta_i| < 1$ (θ_i the eigenvalues) — the interior of the convex hull of the rank-one orthogonal projections.

In Lean (`TraceNormLtOne`): since V carries no inner-product structure in the formalization, self-adjointness with respect to *some* positive-definite g_0 is encoded equivalently as diagonalizability over $\mathbb{R}$ — the structure requires an eigenbasis b with real eigenvalues θ satisfying $\sum_i |\theta_i| < 1$.

Proposition 8 (Parametrization and contractibility; KS Proposition 2.4). $Q_{\mathbb{C}}(V)$ is a fibre bundle over the space of positive-definite inner products on V whose fibre at g_0 is $\Pi(V, g_0)$. Equivalently, choosing a reference inner product,

$$Q_{\mathbb{C}}(V) \cong \text{GL}(V) \times_{\text{O}(V)} \Pi(V).$$

In particular, $Q_{\mathbb{C}}(V)$ is contractible.

Lean status: `True` placeholders (`QC_parametrization`, `QC_contractible`). *Blocked on:* $\text{GL}(V)/\text{O}(V)$ actions and associated-bundle machinery, an inner-product-space structure compatible with the algebraic setup of QC , and a topology on $Q_{\mathbb{C}}(V)$ (needed even to state contractibility).

Proof. (Paper, pp. 10–11.) The diagonalization of Definition 2 presents g as a sequence of angles $\theta_1 \geq \dots \geq \theta_m$ in $(-\pi, \pi)$ and a decomposition $V = V_1 \oplus \dots \oplus V_m$ with $\sum_k \dim V_k \cdot |\theta_k| < \pi$, on V_k the form being $e^{i\theta_k}$ times a positive-definite real form. The subspace $P = \bigoplus e^{-i\theta_k/2} V_k \subset V_{\mathbb{C}}$ is canonical, carries a real positive-definite form g_0 , and $V = \exp(i\pi\Theta/2)(P)$ where Θ is g_0 -self-adjoint with $\|\Theta\|_1 < 1$; condition (4) is exactly the trace-norm bound. $\square$

Definition 9 (Restriction of the form to a subspace). For $g \in Q_{\mathbb{C}}(V)$ and a vector subspace $W \leq V$, the restriction $g|_W: W \times W \rightarrow \mathbb{C}$, $(w_1, w_2) \mapsto g(w_1, w_2)$, is a symmetric $\mathbb{R}$ -bilinear form on W .

In Lean: `AllowableComplexMetric.restrict` (the restricted form) and `AllowableComplexMetric.restrict_sym` (its symmetry), both fully proven.

Proposition 10 (Restriction preserves allowability; KS Proposition 2.5). *If $g \in Q_{\mathbb{C}}(V)$ and W is any vector subspace of V , then $g|_W$ belongs to $Q_{\mathbb{C}}(W)$.*

Lean status: sorry — `restrict_allowable` provides the form and its symmetry (Definition 9); the `nondegenerate` and `angle_cond` fields are sorry'd.

Proof. (Paper, p. 13.) For $g \in Q_{\mathbb{C}}(V)$ the map $v \mapsto \arg g(v)$ is smooth on $\mathbb{P}(V)$ with critical values the angles $\theta_1 \geq \dots \geq \theta_d$, characterized by a min–max over subspaces. It suffices to treat W of codimension one; then the critical values θ'_k of $\arg(g|_W)$ interleave: $\theta_k \geq \theta'_k \geq \theta_{k+1}$, whence $\sum_k |\theta'_k| \leq \sum_k |\theta_k| < \pi$. $\square$

Proposition 11 (Real allowable subspaces of a complex quadratic space; KS Proposition 2.6). *Let Z be a d -dimensional complex vector space with a nondegenerate quadratic form g , and let $\mathcal{R}(Z)$ be the space of d -dimensional real subspaces $A \subset Z$ such that $g|_A$ belongs to $Q_{\mathbb{C}}(A)$ (an open subset of the real Grassmannian). Then $\mathcal{R}(Z)$ is contractible, and*

$$\mathcal{R}(Z) \cong O_{\mathbb{C}}(Z) \times_{O(Z_{\mathbb{R}})} \Pi(Z_{\mathbb{R}}).$$

Not formalized (deferred). *Needed for the Wick rotation of field operators in Section 5 of the paper; planned for `WickRotation/FieldOperators`. Requires the Grassmannian of real subspaces, $O_{\mathbb{C}}(Z)$ -actions and the same associated-bundle machinery as Proposition 8.*

Proof. (Paper, p. 13.) The space $\mathcal{R}(V; Z)$ of $\mathbb{R}$ -linear embeddings $f: V \rightarrow Z$ with $f^*(g)$ allowable is simultaneously a principal $GL(V)$ -bundle over $\mathcal{R}(Z)$ and a principal $O_{\mathbb{C}}(Z)$ -bundle over $Q_{\mathbb{C}}(V)$. The contractibility of $Q_{\mathbb{C}}(V)$ (Proposition 8) and the homotopy equivalence of both structure groups with the compact O_d yield the contractibility of $\mathcal{R}(Z)$. $\square$

Proposition 12 (Domain of holomorphy; KS Proposition 2.7). *The domain $Q_{\mathbb{C}}(V)$ is holomorphically convex, i.e. a domain of holomorphy in $S^2(V_{\mathbb{C}}^*)$.*

Lean status: `True` placeholder (`QC_domain_of_holomorphy`). *Blocked on: complex-manifold structure on the space of quadratic forms, the notion of domain of holomorphy (Stein theory), and Siegel domains — none in Mathlib.*

Proof. (Paper, p. 14.) The Siegel domain $\mathcal{U}(V)$ of complex inner products with positive-definite real part is a domain of holomorphy; so is the product $\prod_{0 \leq p \leq d/2} \mathcal{U}(\Lambda^p(V))$. $Q_{\mathbb{C}}(V)$ is the intersection of this product domain with the affine subvariety given by the natural embedding of $S^2(V_{\mathbb{C}}^*)$ (whose components are the exterior-power forms of Definition 1), hence itself a domain of holomorphy. $\square$

Lemma 13 (Holomorphic extension from an L-shaped tube; KS Lemma 2.8). *Let L be the L-shaped subset $(\{0\} \times [0, 1]) \cup ([0, 1] \times \{0\})$ of the quadrant $(\mathbb{R}_+)^2$. Then any holomorphic function F defined in a neighbourhood of $L \times i\mathbb{R}^2 \subset \mathbb{C}^2$ extends holomorphically to $P \times i\mathbb{R}^2$, where P is the intersection of $(\mathbb{R}_+)^2$ with a neighbourhood of L in $\mathbb{R}^2$.*

Not formalized (deferred with Proposition 6). *Requires several-complex-variables extension theory (continuity-principle / analytic-disc arguments, Bochner tube theorem) not in Mathlib.*

1.3 The Shilov boundary: Lorentzian metrics

Definition 14 (Lorentzian signature). A real symmetric bilinear form φ on V is *Lorentzian* (of signature $(d-1, 1)$) if in some basis it diagonalizes as $\sum_i s_i x_i y_i$ with each $s_i \in \{+1, -1\}$ and exactly one $s_i = -1$.

Proposition 15 (Lorentzian metrics lie on the boundary; KS p. 9). *A real Lorentzian metric φ on V lies on the boundary of $Q_{\mathbb{C}}(V)$: it is a limit of allowable complex metrics (entry-wise on a fixed basis, i.e. in the canonical topology on the finite-dimensional space of forms), but is not itself allowable.*

Lean status: sorry (`lorentzian_on_boundary`).

Proof. Diagonalize $\varphi = \sum_i s_i y_i^2$ with exactly one $s_i = -1$ and perturb the negative coefficient to $e^{i(\pi-\delta)}$: the angle sum is $\pi - \delta < \pi$, so the perturbation is allowable and converges to φ as $\delta \downarrow 0$. That φ itself is not allowable follows from Lemma 4: φ takes a real negative value on the timelike basis vector. $\square$

Proposition 16 (Only Lorentzian metrics lie on the boundary; KS p. 9). *If a real nondegenerate bilinear form φ on V is a limit of allowable complex metrics but is not positive-definite (the positive-definite forms being the real points of the open domain $Q_{\mathbb{C}}(V)$ itself), then φ is Lorentzian. Equivalently (paper): among real nondegenerate metrics, those of signature $(d-1, 1)$ — and no others — lie on the boundary of $Q_{\mathbb{C}}(V)$.*

Lean status: sorry (*only_lorentzian_on_boundary*).

Proof. For a real metric each $|\arg \lambda_i|$ is 0 or π , and inequality (4) shows at most one $|\arg \lambda_i|$ can become π in the limit: a two-dimensional φ -negative-definite subspace would, after restricting the approximating allowable metrics to it (Proposition 10), force the angle sum toward 2π . Nondegeneracy and failure of positive-definiteness then force exactly one negative eigenvalue. $\square$

Theorem 17 (Two copies of the Lorentzian metrics on the Shilov boundary; KS p. 9). *The Shilov boundary of $Q_{\mathbb{C}}(V)$ — regarded as a bounded domain in an affine variety, as in the proof of Proposition 12 — contains two disjoint copies of the space of Lorentzian metrics on V , for an eigenvalue λ can approach the negative real axis either from above or from below. The two copies are interchanged by the complex-conjugation map on $Q_{\mathbb{C}}(V)$. With orientation-reversing elements of $\mathrm{GL}(V)$ acting antilinearly on the orientation line, the nondegenerate points of the Shilov boundary are the time-oriented Lorentzian metrics.*

Lean status: True placeholder (*two_copies_on_boundary*); per the no-approximations policy, no weakened stand-in is stated. The faithful statement is blocked on, in order: (1) induced forms on exterior powers / the Hodge star (the affine-variety embedding $Q_{\mathbb{C}}(V) \rightarrow \prod_{0 \leq p \leq d/2} \mathcal{U}(\Lambda^p V)$ from the proof of Proposition 12, whose $p=0$ factor $(\det g)^{1/2}$ carries the branch that separates the two copies — in the naive closure inside $S^2(V_{\mathbb{C}}^*)$ the copies coincide); (2) Siegel domains and the Cayley transform (the bounded realization); (3) the Shilov boundary notion itself (smallest compact subset of the closure on which every holomorphic function attains its supremum).

1.4 The two-dimensional case

Proposition 18 ($Q_{\mathbb{C}}(V)$ is a polydisc when $\dim V = 2$; KS p. 15). *When $\dim V = 2$, the conformal class $(\det g)^{-1/2}g$ decouples from the volume element, and a nondegenerate complex metric is determined up to scale by its two null directions in $\mathbb{P}(V_{\mathbb{C}})$; allowability puts one in each open hemisphere separated by the equator $\mathbb{P}(V)$. The domain $Q_{\mathbb{C}}(V)$ is thus a 3-dimensional polydisc: one disc for each of the two complex structures (null directions) and one for the volume element.*

Lean status: True placeholder (*QC_two_dim_polydisc*). Blocked on: projective geometry of $\mathbb{P}(V_{\mathbb{C}})$ and the conformal/volume decomposition of quadratic forms.